drmTMB : : CHEAT SHEET

Distributional regression with Template Model Builder

drmTMB 0.7.1

MODEL SPECIFICATION

drm_formula() / bf()	Combine one formula per distributional parameter. bf() is the short alias.
drmTMB()	Fit a univariate or bivariate distributional model. The native TMB engine is the default.
cmr_data()	Validate encounter histories and keyed occasion or interval data for the dedicated CJS route.
drm_cjs()	Fit first-capture-conditional apparent survival and detection; this route does not estimate abundance.
drm_control()	Set optimizer controls and fitted-object storage or large-data options.
objective_at()	Evaluate a fitted model's objective at a supplied parameter point, without refitting -- pairs with drm_control(start = ...) to compare a fit against another engine's answer.
drm_provenance()	Report which BUILD produced a fit -- git SHA, dirty flag and build time -- when the package version alone cannot tell two builds apart.
meta_V()	Supply known sampling variance or covariance for Gaussian meta-analytic models.
meta_vcov_bivariate()	Assemble the row-paired known covariance matrix for two-response meta-analysis.
miss_control()	Choose the documented missing-response or missing-predictor policy.
mi()	Mark a predictor whose missing values are modelled jointly with the response.
impute_model()	Specify the family and formula for a modelled missing predictor.

PAIRED-OUTCOME ASSOCIATION

biv_associate()	Fit two margins for a pair class listed in the association guide, then construct their frozen-margin association.
associate_pairs()	Estimate within-row association from two fitted, aligned margins on a route listed in the association guide.
latent_normal()	Declare the Gaussian-copula kernel used by the paired-outcome association stage.

RESPONSE AND PREDICTOR FAMILIES

student()	Student-t location-scale-shape models for heavy-tailed continuous responses.
skew_normal()	Skew-normal location-scale-shape models.
lognormal()	Positive continuous responses on the lognormal scale.
tweedie()	Non-negative semicontinuous responses that may include exact zeroes.
beta_family()	Strict proportions inside (0, 1).
zero_one_beta()	Continuous proportions with structural exact zeroes or ones.
beta_binomial()	Success counts with a known denominator and extra-binomial variation.
categorical()	An unordered categorical missing-predictor model in impute_model().
cumulative_logit()	Ordered categorical responses or ordered missing-predictor models.
nbinom2()	Negative-binomial count models with quadratic mean-variance scaling.
truncated_nbinom2()	Positive or hurdle negative-binomial count models.
biv_gaussian()	Two Gaussian responses with separate means, scales, and residual rho12.
biv_lognormal()	Two lognormal responses with residual rho12 on the log-response scale.
biv_student()	Two heavy-tailed Student-t responses with a shared degrees-of-freedom parameter.
gaussian(), Gamma(), binomial(), poisson()	Base-R families for documented Gaussian, positive, event-count, and count-response routes.

RANDOM AND STRUCTURED EFFECTS

animal()	Pedigree or supplied animal-model precision/covariance structure.
phylo()	A structured effect derived from one phylogenetic tree.
phylo_interaction()	A pair-level effect derived from two phylogenetic trees.
spatial()	Coordinate-based spatial structured effects.
spatial_coords()	Convert longitude and latitude to an explicitly chosen projected coordinate system.
make_mesh()	Build the fixed-kappa SPDE mesh and projection inputs used by mesh spatial effects.
relmat()	A structured effect from a user-supplied relationship or precision matrix.
corpair()	Mark one current q=2 labelled ordinary or phylogenetic location-location correlation formula. Keep this distinct from residual rho12 = ~ ....
sd(group) ~ ..., sd1(group) ~ ..., sd2(group) ~ ...	Formula-only syntax for Gaussian location random-effect SDs. These are not replacements for stats::sd().
drm_phylo_penalty()	Define the documented penalized/MAP prior for weakly identified phylogenetic SDs or correlations. Penalized fits are MAP fits, not ordinary ML fits for LR tests or AIC comparisons.
drm_phylo_penalty_sweep()	Check whether a penalized phylogenetic coupling is data-informed or prior-shaped.

DISTRIBUTIONAL ASSESSMENT

fitted_distribution()	Access fitted per-row density, CDF, and quantile functions.
exceedance()	Compute conditional probability above or below a threshold.
centile_chart()	Plot conditional response centiles over one covariate.
qq_plot()	Plot randomized quantile residuals against normal quantiles.
worm_plot()	Detrend the QQ plot to reveal systematic departures.

Deprecated compatibility markers omitted. Navigation guide, not a capability claim.

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CHECK AND EXTRACT - CORE

check_drm()	Run the model-aware convergence and identifiability checks. Start here after fitting.
convergence_status()	Return "converged", "boundary", or "degenerate" when a logical flag is not enough.
is_converged()	Return one logical convergence summary for programmatic workflows.
summary()	Print coefficients, fit information, and optional intervals.
coef() / fixef()	Extract fixed-effect coefficient blocks; fixef() makes the mixed-model meaning explicit.
ranef()	Extract conditional modes for ordinary or structured random effects.
sigma()	Extract fitted residual or family-specific scale values.
rho12()	Extract within-observation residual correlation for bivariate Gaussian fits.
corpairs()	Extract labelled ordinary or structured latent-correlation summaries.
structured_effects()	List fitted structured-effect terms and their model roles.
heritability() / icc() / repeatability()	Report the share of total variance carried by one structured random-effect component, on the working (log-SD) scale. The three differ only in their denominator.

FORMULA ANATOMY

drm_formula(mu = trait ~ temp, sigma = ~ temp)	Model the expected trait and its residual scale with separate linear predictors.
nu = ~ 1, zi = ~ habitat, hu = ~ habitat	Add a shape, zero-inflation, or hurdle component where the chosen family supports it.
rho12 = ~ environment	Model coscale: the within-observation residual correlation of a bivariate Gaussian fit.
sd(group) ~ x, sd1(group) ~ x, sd2(group) ~ x	Model Gaussian location random-effect SDs; use level = "phylogenetic" for the documented structured route.
corpair(label) ~ x	Formula-marker syntax for a current q=2 labelled ordinary or phylogenetic location-location correlation block; predictors must be constant within group.

CHECK AND EXTRACT - FIT

coevolution_cor() / coevolution_vc() / coevolution_summary()	Read the among-axis structure of a q = 4 coevolution fit, where mu1, mu2, sigma1 and sigma2 share one structured effect with a dense 4 x 4 among-axis covariance.
imputed()	Summarize fitted values for a modelled missing predictor.
fitted()	Extract fitted response means for the original fitted rows.
residuals()	Compute the documented response, Pearson, or randomized-quantile residuals.
weights()	Recover fitted likelihood weights.
logLik()	Extract the unpenalized fitted log likelihood.
AIC() / BIC()	Compare appropriate maximum-likelihood fits using information criteria.
aicc()	Second-order small-sample correction to AIC(). Prefer it when the fitted-row count is small relative to the number of parameters.
vcov()	Extract the fixed-effect covariance matrix when the Hessian is usable.
nobs() / df.residual() / deviance()	Extract fitted-row count, residual degrees of freedom, or absolute negative twice log likelihood.

QUICK ECOLOGICAL RECIPES

drmTMB(drm_formula(trait ~ temp, sigma = ~ temp), family = gaussian(), data = dat)	Ask whether both the expected trait and residual variability change with temperature.
drmTMB(count ~ habitat, family = nbinom2(), data = dat)	Model overdispersed abundance counts across habitats.
drmTMB(cbind(success, failure) ~ treatment, family = beta_binomial(), data = dat)	Model success probability with extra-binomial variation.
check_drm(fit); predict(fit, newdata = grid)	Check the fitted object before interpreting predictions on a controlled grid.
profile_targets(fit); confint(fit, method = "profile", parm = target)	Discover a direct target before requesting a profile-likelihood interval.

INTERVALS AND PROFILES

profile_targets()	Discover direct fitted targets and their profile names before requesting an interval.
confint()	Compute available Wald, profile-likelihood, or parametric-bootstrap intervals.
profile()	Compute likelihood-ratio profiles for explicit targets.
plot() for a profile	Plot the likelihood-ratio curve, cutoff, estimate, and interval endpoints.
chibar_pvalue() / lrt_boundary()	Test q variance components = 0 against the correct chi-bar-square mixture. The ordinary chi-squared reference is wrong at a boundary and conservative, so it loses power.

PREDICT, SIMULATE, PLOT

prediction_grid()	Build a new-data grid that varies selected predictors while controlling the others.
predict()	Predict the response or a named distributional parameter for original or new data.
predict_parameters()	Return several fitted distributional parameters in one tidy long table.
marginal_parameters()	Compute unweighted plug-in averages of fitted parameters on their moment-appropriate reporting scales; this is not an interval or contrast.
simulate()	Simulate new responses from a fitted model.
plot_corpairs()	Plot a table returned by corpairs().
plot_parameter_surface()	Plot a table returned by predict_parameters() or a compatible parameter-surface table.

INTERPRETATION GUARDRAILS

student() scale	Student-t sigma is a scale parameter, not the response standard deviation.
drm_phylo_penalty() fits	These are penalized/MAP fits; logLik() reports the unpenalized data likelihood, and ordinary LR/AIC comparisons are not standard.
Capability matrix	A visible function does not imply every family-by-structure interval route is